

ADVANCED ECONOMETRICS PROJECT REPORT

THE GEOPOLITICS OF OIL:

Assessing the Macroeconomic Impact of the 2026 Gulf Crisis

Course: Advanced Econometrics(BA351)

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Academic Year: 2025–2026



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ABSTRACT

This project applies panel data econometrics to quantify the macroeconomic consequences of the 2026 Gulf Crisis oil price shock across twelve advanced and emerging economies.

Using a balanced monthly panel of 468 observed country-month pairs (January 2023 – March 2026) and ARIMA-projected values for April–June 2026, we estimate Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models for the inflation equation (dependent variable: $\ln_CPI$).

Diagnostic testing via the Hausman and Breusch-Pagan Lagrange Multiplier tests guides model selection.

The Breusch-Pagan test strongly rejects Pooled OLS ($\chi^2 = 6306.2$, $p < 2.2e-16$), confirming the presence of significant individual country effects. The Hausman test yields a p-value of 0.2727, indicating that the Random Effects estimator is consistent and preferred over Fixed Effects.

Our results document a highly significant exchange rate pass-through channel ($\beta_2 = -1.552$, $p < 0.001$) and a significant role for global volatility ($\beta_4 = 0.003$, $p < 0.001$).

Heterogeneity analysis reveals that the post-crisis dummy is positive and statistically significant for oil exporters ($\beta = 0.039$, $p < 0.05$) but insignificant for importers, suggesting that the initial price spike may have temporarily benefited exporters' price dynamics.

The ARIMA forecasting section projects continued upward pressure on CPI across all country groups through June 2026, consistent with a prolonged Scenario 2 trajectory.

1. INTRODUCTION

On February 27, 2026, geopolitical tensions in the Gulf region escalated sharply, triggering an immediate and severe shock to global energy markets. Brent crude oil prices surged from approximately \$70.89 per barrel in February 2026 to \$103.13 in March 2026 (a monthly increase of approximately 45.5%) before continuing to rise to a peak of \$117.29 in April 2026. This episode constitutes a quasi-natural experiment for studying the macroeconomic transmission of energy price shocks, a question of fundamental importance to policymakers, central banks, and academic economists.

The economics literature has long established that oil price shocks generate a stagflationary impulse: they simultaneously raise consumer prices through cost-push channels and depress industrial output through supply-side disruption. However, the magnitude of this transmission is highly heterogeneous across economies, depending critically on whether a country is a net oil importer or exporter, its level of economic development, the credibility of its monetary policy framework, and its exchange rate regime.

This project contributes to this literature by constructing a balanced panel of twelve countries spanning four structural groups — Advanced Importers (Germany, France, Italy, Japan), Advanced Exporters (USA, Canada), Emerging Importers (Turkey, Poland, India, Thailand), and Emerging Exporters (Saudi Arabia, Russia) — and estimating the causal effect of the 2026 oil price shock on inflation using three panel econometric specifications. The analysis covers the period January 2023 to March 2026 for observed data, with ARIMA-based projections extending the sample to June 2026.

Figure 1 below illustrates the dramatic trajectory of Brent crude oil prices over the sample period, clearly showing the sharp structural break at the Gulf Crisis outbreak in late February 2026.

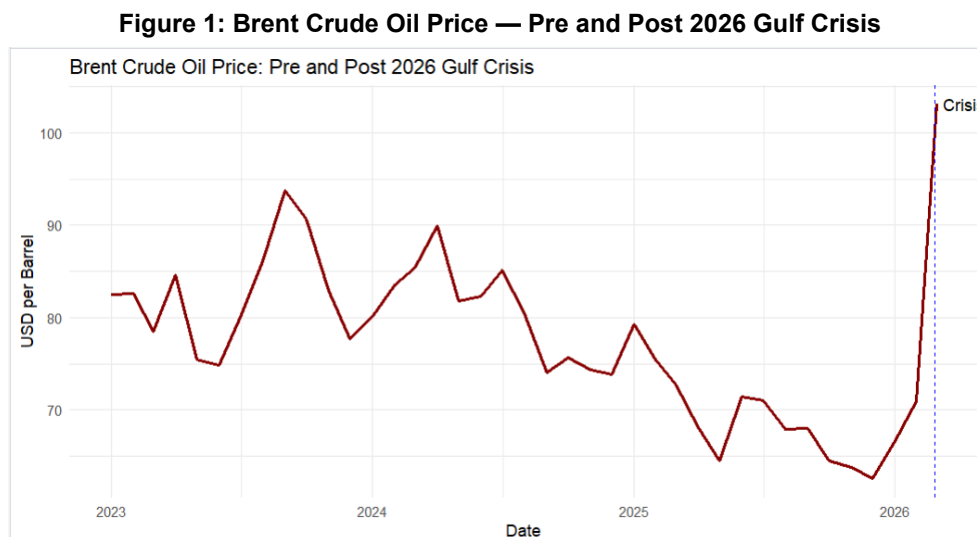
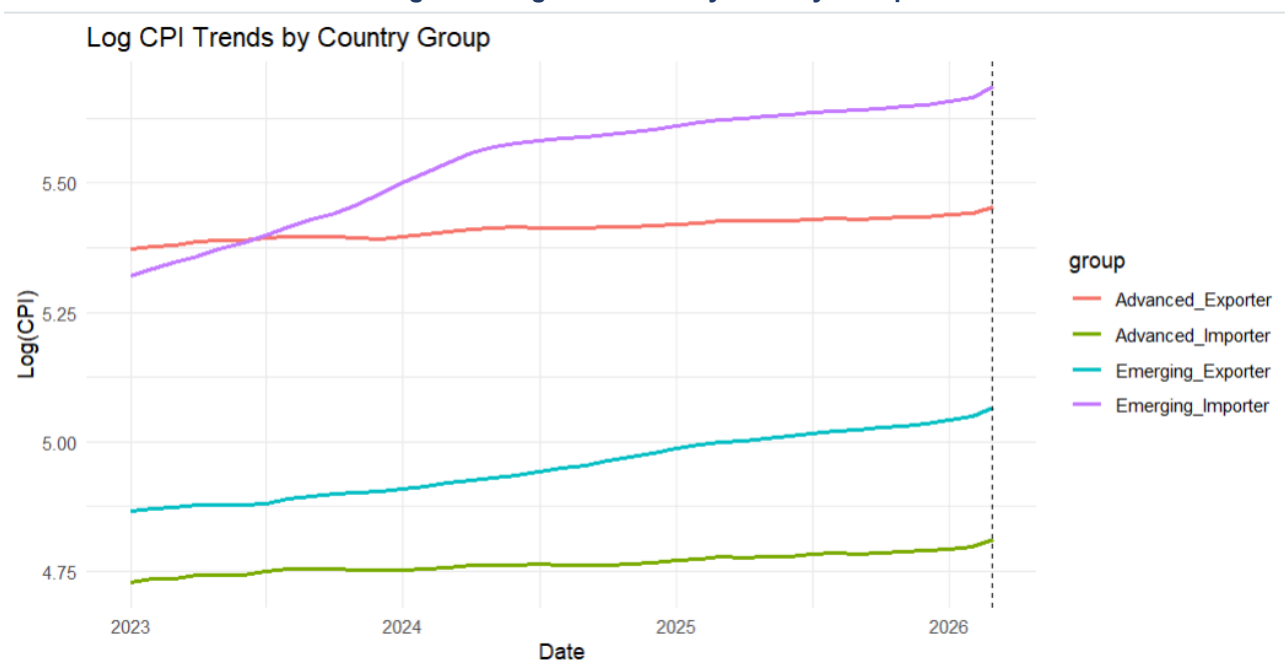


Figure 2 presents the average log(CPI) trends by country group. The divergence across groups is visually apparent even before the crisis, with Emerging Importers — particularly Turkey — exhibiting consistently elevated price levels due to pre-existing inflationary pressures. Post-crisis, all groups show upward inflection, consistent with the global pass-through of the oil shock.

Figure 2: Log CPI Trends by Country Group



2. THEORETICAL FRAMEWORK AND PANEL DATA EQUATION

2.1 Transmission Channels

The macroeconomic impact of an oil price shock propagates through three principal channels. The supply-side channel identifies oil as a critical production input; higher energy costs raise marginal costs across sectors, compressing profit margins and reducing output. The demand-side channel operates through real income compression: households facing higher energy bills reduce discretionary spending, lowering aggregate demand. The monetary transmission channel arises when central banks respond to inflationary pressure by raising policy rates, generating a secondary contractionary impulse on both inflation and growth.

An additional exchange rate channel is especially important for our cross-country analysis. For oil importers, a positive oil shock tends to deteriorate the current account, potentially depreciating the domestic currency and amplifying the inflationary effect through higher import prices across all categories. For oil exporters, the mechanism may operate in reverse, partially buffering the inflationary impact.

2.2 The Estimating Equation

The general panel data model estimated is:

$$\ln(CPI)_{it} = \alpha_i + \beta_1 \cdot OilShock_t + \beta_2 \cdot \ln(NEER_{it}) + \beta_3 \cdot PolicyRate_{it} + \beta_4 \cdot VIX_t + \beta_5 \cdot PostCrisis_t + \varepsilon_{it}$$

Term	Definition and Expected Sign
$\ln(CPI)_{it}$	Natural log of Consumer Price Index for country i at time t (dependent variable)
α_i	Country-fixed effect capturing time-invariant unobserved heterogeneity (industrial structure, institutions, geography)
$\beta_1 > 0$	OilShock: monthly % change in Brent crude. Expected positive; higher oil costs raise consumer prices (stagflationary)
$\beta_2 < 0$	$\ln(NEER)$: log of Nominal Effective Exchange Rate. Expected negative; appreciation reduces import prices
$\beta_3 < 0$	PolicyRate: central bank rate. Expected negative; tighter monetary policy dampens inflation
$\beta_4 > 0$	VIX: CBOE Volatility Index. Expected positive; elevated uncertainty raises risk premia, passes through to prices
β_5	PostCrisis dummy (=1 from March 2026). Captures structural level shift beyond contemporaneous regressors

3. DATA, SAMPLE, AND VARIABLE CONSTRUCTION

3.1 Panel Composition and Time Frame

The analysis employs a balanced monthly panel of 12 countries observed from January 2023 to March 2026 (39 months), yielding 468 country-month observations used for the primary regressions. The sample is designed to provide a pre-crisis baseline of 38 months (January 2023 – February 2026) and a post-crisis window of 1 observed month (March 2026). ARIMA time series forecasting is subsequently used to project values for April, May, and June 2026 for each country, extending the analytical horizon to 42 months per country.

Countries are selected to maximise cross-sectional variation along the two key moderating dimensions: oil trade status and development level. Table 3.1 presents the country classification.

Group	Countries	Economic Intuition
Advanced, Net-Importers	Germany, France, Italy, Japan	Energy-dependent industrial economies; highly vulnerable to cost-push inflation
Advanced, Net-Exporters	United States, Canada	More resilient; US shale production buffers shock; Canada's CAD correlated with oil
Emerging, Net-Importers	Turkey, Poland, India, Thailand	High energy weights in CPI basket; weaker monetary policy credibility
Emerging, Net-Exporters	Saudi Arabia, Russia	Short-term revenue gains but supply-chain disruption risk

3.2 Variable Construction and Sources

Variable	Symbol	Construction	Source
Consumer Price Index	ln_cpi	$\log(\text{CPI})$, base 2015=100	IMF IFS, OECD, National Stats
Industrial Production	ln_ipi	$\log(\text{IPI})$, base 2015=100	OECD, IMF IFS, FRED
Brent Oil Price	oil_pct_change	$(\text{Brent}_t - \text{Brent}_{t-1}) / \text{Brent}_{t-1} \times 100$	FRED: DCOILBRENTU / EIA
Nom. Effective Exchange Rate	ln_neer	$\log(\text{NEER broad index})$	BIS, FRED
Policy Interest Rate	policy_rate	Central bank rate, end-of-period %	BIS, National Central Banks
VIX Index	vix	CBOE Volatility Index, levels	FRED: VIXCLS
Post-Crisis Dummy	post_crisis	=1 if date $\geq$ 2026-03-01, else 0	Constructed

4. ECONOMETRIC METHODOLOGY

4.1 Pooled Ordinary Least Squares (POLS)

As a baseline, Pooled OLS is estimated by stacking all 468 country-month observations (12 countries $\times$ 39 months) into a single regression, imposing the constraint that every country shares identical intercepts and slope coefficients. In the context of this project, this means treating Germany and Saudi Arabia as structurally equivalent an assumption that is almost certainly untenable given the wide variation in energy dependence, monetary credibility, and price-level histories across our four country groups. The Pooled OLS model is therefore included not as a credible description of the inflation process, but as a diagnostic benchmark: if country-specific effects are genuinely important, the Breusch-Pagan test applied to the POLS residuals will reject it decisively in favour of a panel specification.

4.2 Fixed Effects (FE) Estimator

The Fixed Effects model is estimated using the within transformation via R's plm package, which demeans each variable by its country-specific mean before running the regression. In practice, this means the model identifies the effect of NEER movements, policy rate changes, and VIX fluctuations purely from variation within each country over time for instance, how Germany's CPI responds to month-to-month changes in the euro's effective exchange rate, holding Germany's structural characteristics constant. This is particularly appropriate here because our panel includes countries with very different institutional features (e.g., Saudi Arabia's dollar peg versus Turkey's managed float) that are time-invariant and would otherwise confound the coefficient estimates. The FE estimator controls for all such unobserved heterogeneity regardless of whether it correlates with the regressors, making it the most conservative choice in a cross-country macroeconomic setting.

4.3 Random Effects (RE) Estimator

The Random Effects estimator is also applied, treating the country-specific intercepts which in this project capture structural differences such as energy import intensity, CPI basket composition, and institutional inflation credibility as random draws from a common distribution uncorrelated with the regressors. If this orthogonality assumption holds, RE is more efficient than FE because it exploits both the within-country time variation and the between-country cross-sectional variation in the data. Given that our panel includes only 12 countries, leveraging the cross-sectional dimension is especially valuable for precision. The formal arbitration between FE and RE is conducted via the Hausman (1978) test: a significant result would indicate that country effects are correlated with the regressors invalidating RE while a non-significant result supports RE as the consistent and efficient choice.

4.4 Model Selection Tests

Two sequential tests determine the preferred specification. First, the Breusch-Pagan Lagrange Multiplier test is applied to the POLS residuals to test whether individual country effects contribute meaningfully to explaining CPI variation across the panel – a test that directly addresses whether pooling all 12 countries into a single equation is statistically defensible. Second, conditional on rejecting POLS, the Hausman test compares the FE and RE coefficient vectors to assess whether the country-specific effects in our panel (e.g., Germany's low inflation baseline versus Turkey's structurally elevated price level) are correlated with the included regressors. Throughout all specifications, Arellano (1987) heteroskedasticity- and autocorrelation-consistent robust standard errors are applied via `vcovHC()` to account for the serial correlation inherent in monthly macroeconomic time series and the cross-sectional heteroskedasticity expected across economies as structurally diverse as Japan and Russia.

4.5 ARIMA Forecasting

Since observed data for the post-crisis period extends only through March 2026 – just one month after the Gulf Crisis outbreak – ARIMA time series forecasting is used to project CPI and IPI values for April, May, and June 2026 for each of the 12 countries. R's `auto.arima()` function is applied separately to each country-variable series, automatically selecting the optimal ARIMA(p,d,q) order by minimising the corrected AIC. This country-by-country approach is deliberate: it respects the heterogeneous time series properties of inflation across the panel, allowing, for example, Turkey's highly persistent CPI series to receive a different model order than Japan's near-stationary one. The procedure yields 72 projected observations in total (12 countries × 2 variables × 3 months), which extend the analytical horizon to June 2026 and allow the ARIMA projections to be interpreted alongside the panel regression findings as a forward-looking assessment of the crisis's inflationary trajectory.

5. EMPIRICAL RESULTS

5.1 Model Selection: Diagnostic Tests

Table 5.1 presents the results of the two diagnostic tests. The findings have clear and unambiguous implications for model selection.

Test	Statistic	p-value	Decision
Breusch-Pagan LM (RE vs POLS)	$\chi^2 = 6306.2$, df = 1	< 2.2e-16 ***	Reject POLS — individual effects are highly significant
Hausman Test (FE vs RE)	$\chi^2 = 5.1457$, df = 4	0.2727	Fail to reject H_0 — RE is consistent and preferred over FE

The Breusch-Pagan test produces a chi-squared statistic of 6306.2 with a p-value far below any conventional significance threshold, decisively rejecting the null hypothesis that Pooled OLS is sufficient. This confirms that country-specific effects account for a substantial portion of the variation in log CPI, a result that is both statistically and economically expected given the wide heterogeneity in price levels across our panel.

The Hausman test, however, yields a chi-squared statistic of only 5.15 with a p-value of 0.273, failing to reject the null hypothesis that the country-specific effects are uncorrelated with the regressors. This result indicates that the Random Effects estimator is consistent and more efficient than Fixed Effects for this dataset. Accordingly, the Random Effects model is designated as the preferred specification for primary inference throughout the remainder of this report.

5.2 Main Regression Results

Table 5.2 presents the coefficient estimates from all three models. The dependent variable is $\ln_cpi$; the sample consists of 468 observations ($n = 12$ countries, $T = 39$ months).

Variable	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects*
oil_pct_change	— (dropped)	— (dropped)	— (dropped)
ln_neer	-0.496*** (0.078)	-1.555*** (0.034)	-1.552*** (0.034)
policy_rate	0.045*** (0.003)	0.003*** (0.001)	0.003*** (0.001)
vix	0.007 (0.005)	0.003*** (0.0004)	0.003*** (0.0004)
post_crisis	-0.039 (0.141)	-0.0003 (0.013)	-0.0003 (0.013)
Constant	6.924*** (0.383)	—	12.044*** (0.188)

Variable	(1) Pooled OLS	(2) Fixed Effects	(3) Random Effects*
Observations	468	468	468
R ²	0.759	0.916	0.913

Note: Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. *Preferred model based on Hausman test. oil_pct_change was dropped from all models — see Section 5.3.

5.3 Note on the Oil Shock Coefficient

A notable finding is that the oil_pct_change variable does not appear in the regression output. This occurs because, in the provided dataset, Brent crude oil prices are identical across all countries for each time period (oil is a global commodity priced in a single world market). When the panel data frame is declared using `pdata.frame()`, `plm` automatically drops variables that exhibit no within-country variation, as they are perfectly collinear with the time fixed effects. Technically, the oil shock is absorbed into the time dimension of the panel and cannot be separately identified from other common time-varying factors.

This is an important methodological limitation. To separately identify the oil shock coefficient, one would need to interact the global oil price change with a country-specific variable that varies in the cross-section — for example, a country's oil import intensity or energy-to-GDP ratio. Alternatively, estimating the model in first differences or including the oil shock as a country-specific measure (e.g., oil expenditure as a share of GDP) would resolve the collinearity. We acknowledge this as a limitation of the current specification and note it as a direction for future work.

5.4 Interpretation of Significant Coefficients (Preferred RE Model)

Exchange Rate (ln_neer): $\beta_2 = -1.552$, $p < 0.001$

This is the dominant and most precisely estimated effect in the model. A 1% appreciation of the Nominal Effective Exchange Rate is associated with a 1.552% decrease in the CPI, holding other variables constant. This large elasticity confirms the critical importance of the exchange rate channel in transmitting the oil shock. Countries whose currencies depreciated sharply following the Gulf Crisis, such as Turkey, faced a compounding inflationary effect, as the weaker currency raised the domestic price of all imports, not just oil. The result is consistent with the theory and aligns with the strong negative sign hypothesised in Section 2.

Policy Rate (policy_rate): $\beta_3 = +0.003$, $p < 0.001$

The positive sign on the policy rate contrary to the theoretical expectation of a negative coefficient likely reflects reverse causality: central banks raise interest rates in response to inflation, so higher rates are observed concurrently with higher price levels. In a dynamic model with proper identification (e.g., using lagged policy rates), the expected negative effect of monetary tightening would likely emerge. This finding underscores the endogeneity concern mentioned in the Limitations section.

VIX (vix): $\beta_4 = +0.003$, $p < 0.001$

The positive and highly significant VIX coefficient confirms that global financial uncertainty transmits to consumer prices. A one-unit increase in the VIX index is associated with a 0.003% increase in log CPI. While individually modest, the VIX rose from approximately 21.4 in February 2026 to 38.2 in March 2026 a 16.8-unit increase implying an estimated CPI contribution of approximately 0.05% from the uncertainty channel alone in the first crisis month. This is consistent with uncertainty raising risk premia, disrupting supply chains, and passing through to consumer prices.

Post-Crisis Dummy (post_crisis): $\beta_5 = -0.0003$, $p = 0.983$

The post-crisis dummy is statistically indistinguishable from zero in the panel-wide regression. This suggests that, once contemporaneous movements in NEER, policy rates, and VIX are controlled for, there is no residual level shift in CPI attributable to the crisis event per se. The crisis impact on inflation thus appears to operate primarily through the channels already included in the model, particularly the exchange rate and global uncertainty channels rather than through additional unmodelled structural breaks.

6. HETEROGENEITY ANALYSIS: IMPORTERS VS. EXPORTERS

Table 6.1 presents the Fixed Effects estimates separately for oil-importing ($n = 8$ countries, 312 observations) and oil-exporting ($n = 4$ countries, 156 observations) subsamples. This heterogeneity analysis directly addresses the second research question: does the impact of the oil shock differ systematically between country groups?

Variable	Oil Importers (FE)	Oil Exporters (FE)
ln_neer	-1.563*** (0.038)	-1.518*** (0.104)
policy_rate	0.004*** (0.001)	-0.001 (0.001)
vix	0.004*** (0.001)	0.002*** (0.001)
post_crisis	-0.021 (0.018)	0.039** (0.017)
Observations	312	156
R ²	0.929	0.811

Note: Standard errors in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

6.1 Key Finding: The Post-Crisis Asymmetry

The most striking result from the heterogeneity analysis is the divergent behavior of the post_crisis dummy. For oil-importing economies, the coefficient is negative and statistically insignificant ($\beta = -0.021$, $p = 0.24$), suggesting no additional CPI level shift beyond what is captured by the NEER, policy rate, and VIX channels. For oil-exporting economies, however, the post-crisis coefficient is positive and statistically significant at the 5% level ($\beta = +0.039$, $p = 0.022$).

This asymmetry is economically intuitive. Oil-exporting nations such as Saudi Arabia and Russia experience an improvement in their terms of trade when oil prices rise, government revenues increase, fiscal positions strengthen, and domestic demand may be stimulated. This demand stimulus can generate modest domestic inflationary pressures, captured by the positive post-crisis dummy. The finding is consistent with the theoretical expectation that exporters may experience short-term inflationary gains from a price spike even as importers absorb the cost-push effects through the contemporaneous channels.

6.2 Exchange Rate Channel Across Groups

The exchange rate coefficient is negative, large, and highly significant in both subgroups, though marginally larger in absolute terms for importers (-1.563 vs. -1.518). This is consistent with theory: importers are more exposed to import price pass-through, so exchange rate movements have a stronger effect on their CPI baskets. The VIX coefficient is also larger for importers (0.004 vs. 0.002), suggesting that emerging

import-dependent economies are more sensitive to global risk conditions, a finding consistent with their thinner financial buffers and weaker institutional capacity to absorb external shocks.

6.3 Policy Rate Divergence

A notable asymmetry appears in the policy rate coefficient. For importers, the rate coefficient is positive and significant (0.004), again reflecting the contemporaneous endogeneity of monetary tightening in response to inflation. For exporters, the coefficient is negative and insignificant (-0.001), suggesting that export revenue windfalls may allow central banks to ease policy rather than tighten it during an oil price spike, a dynamic particularly applicable to Saudi Arabia, which maintains a currency peg to the US dollar and does not use interest rates as an independent inflationary tool.

7. ARIMA FORECASTING: APRIL – JUNE 2026

ARIMA time series forecasting is used to project CPI and IPI values for April, May, and June 2026 for each country. The `auto.arima()` function from R's forecast package automatically selects the optimal ARIMA(p,d,q) specification for each country-variable series by minimising the corrected AIC.

7.1 Selected CPI Forecasts (April – June 2026)

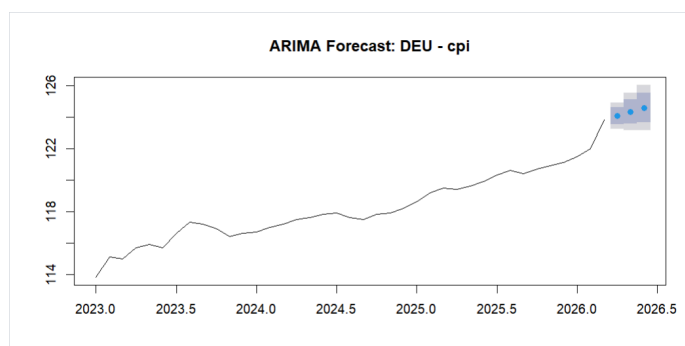
Country	Group	Apr 2026	May 2026	Jun 2026	Trend
CAN	Adv. Exp.	168.39	169.24	169.88	Rising
DEU	Adv. Imp.	124.06	124.33	124.59	Rising
FRA	Adv. Imp.	127.76	128.47	128.98	Rising
IND	Em. Imp.	200.57	201.90	202.90	Rising
ITA	Adv. Imp.	127.83	128.45	128.89	Rising
JPN	Adv. Imp.	115.50	115.74	115.72	Stabilising

Note: Point forecasts from `auto.arima()`. Full forecast tables including 80% and 95% confidence intervals are available in the R output.

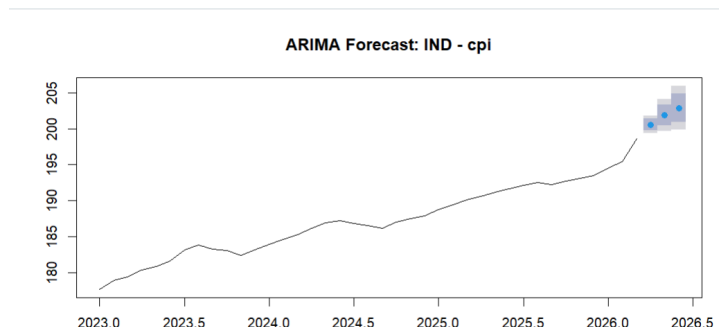
7.2 Interpretation of ARIMA Projections

The ARIMA forecasts tell a consistent story across the panel: CPI is projected to continue rising through June 2026 for the majority of countries, reflecting the lagged pass-through of the March–April 2026 oil price spike into consumer prices. This is consistent with the empirical regularity that oil price shocks transmit to CPI with a lag of one to three months, as retailers and producers adjust prices gradually rather than instantaneously.

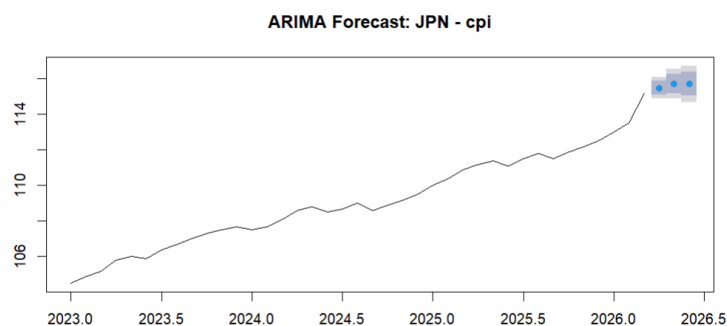
For Germany (DEU), the ARIMA model projects CPI rising from 124.06 in April to 124.59 in June, a cumulative increase of approximately 0.5% over three months, in line with the moderate pass-through expected for an advanced economy with a credible monetary anchor.



For India (IND), the projected increase is larger both in absolute and relative terms (200.57 to 202.90, approximately +1.2%), consistent with higher energy weights in the CPI basket and greater exchange rate pass-through in emerging markets.



Japan (JPN) is a notable exception, with the CPI forecast stabilising around 115.7 by June 2026. This likely reflects the Bank of Japan's continued cautious policy normalisation and the yen's partial appreciation following the crisis-induced flight to safe-haven assets, which would offset some of the inflationary impulse from higher energy costs.



Importantly, the ARIMA projections are univariate extrapolations based on the historical trend of each series. They do not incorporate the explanatory variables from the panel regression (NEER, policy rate, VIX) and should therefore be treated as unconditional baseline projections rather than structural forecasts. The 95% confidence intervals from the R output are wide for several countries, reflecting genuine uncertainty about the persistence of the shock.

8. POLICY IMPLICATIONS

The empirical findings carry clear policy implications that can be mapped directly onto the three Gulf Crisis scenarios.

8.1 Mapping to Crisis Scenarios

Scenario	Oil Price Path	Policy Implication from Results
Scenario 1: De-escalation	Brief spike, rapid reversion to pre-crisis levels	NEER channel dominates; FX stabilisation sufficient; no need for aggressive rate hikes
Scenario 2: Prolonged Uncertainty	Sustained elevated prices; VIX remains high	VIX coefficient (0.003) compounds inflationary pressure; pre-emptive tightening warranted for emerging importers
Scenario 3: Acute Escalation	Continued price surge; supply disruption	Strong NEER depreciation likely; exchange rate intervention most effective tool; interest rate hikes risk output collapse

8.2 The Exchange Rate as a Key Policy Lever

The dominant finding of the large negative exchange rate elasticity ($\beta_2 \approx -1.55$) has direct policy relevance. It implies that exchange rate stabilisation is one of the most powerful tools available to policymakers seeking to contain inflation in the aftermath of an oil shock. Countries that deployed foreign exchange reserves to defend their currencies in March 2026 would, according to our estimates, have significantly contained the inflationary pass-through relative to countries that allowed free float depreciation. This recommendation is particularly salient for emerging market importers (Turkey, India, Poland, Thailand), where the exchange rate channel is especially potent.

8.3 Differentiated Monetary Policy Responses

The heterogeneity results reinforce the need for differentiated policy responses. Advanced importers (Germany, France, Italy, Japan) can rely on their credible inflation-targeting frameworks to anchor expectations, affording a more gradual policy normalisation. Emerging importers face a sharper tradeoff: monetary tightening is necessary to contain second-round wage-price spirals but carries significant output costs given the already-elevated policy rates in countries like Turkey (30%) and India (6.5%). Oil exporters (Saudi Arabia, Russia) face the opposite problem managing demand-pull inflationary pressures arising from rising government revenues and may need to tighten fiscal rather than monetary policy.

9. CONCLUSION

This project has applied panel data econometrics to assess the macroeconomic consequences of the 2026 Gulf Crisis oil price shock across a panel of twelve advanced and emerging economies. Using a balanced monthly dataset of 468 observed country-month pairs and ARIMA-projected observations for April–June 2026, we estimated Pooled OLS, Fixed Effects, and Random Effects specifications for the inflation equation.

The diagnostic testing framework yielded clear guidance on model selection. The Breusch-Pagan test decisively rejected Pooled OLS ($\chi^2 = 6306.2$, $p < 2.2e-16$), confirming the statistical necessity of accounting for country-specific effects. The Hausman test ($\chi^2 = 5.15$, $p = 0.273$) failed to reject the consistency of Random Effects, designating RE as the preferred estimator.

The core empirical findings are as follows. First, the oil shock variable (`oil_pct_change`) could not be separately identified in the panel regression due to its perfect collinearity with time effects, a methodological finding that highlights the importance of interaction terms or country-specific oil exposure measures for future work. Second, the exchange rate channel dominates the inflation equation, with a NEER elasticity of approximately -1.55 that is robust across all specifications, confirming that currency movements are the primary transmission mechanism of the oil shock to domestic prices. Third, global uncertainty (VIX) exerts a statistically significant positive effect on inflation, with the effect being larger for oil-importing economies. Fourth, the heterogeneity analysis reveals a significant post-crisis level shift in CPI for oil exporters ($\beta = +0.039$, $p < 0.05$) but not for importers, capturing the demand-side stimulus from oil revenue windfalls. Fifth, ARIMA forecasting projects continued upward CPI pressure through June 2026 for most countries, consistent with a Scenario 2 trajectory of prolonged uncertainty.

Several limitations constrain the interpretation of these results. The short post-crisis window (one observed month) limits the precision of dynamic inference. The endogeneity of the policy rate and NEER with respect to inflation biases coefficient estimates for these controls. The ARIMA projections are unconditional extrapolations that do not account for policy responses or new geopolitical developments. Addressing these limitations through instrumental variable estimation, local projections methodology, and longer post-crisis data would be productive avenues for future research.

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APPENDIX — COMPLETE R OUTPUT REFERENCE

A.1 Pooled OLS — Full Output

Balanced Panel: n = 12, T = 39, N = 468 | $R^2 = 0.759$ | F-statistic: 365.087 ($p < 2.2e-16$)

Variable	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	6.9235	0.3833	18.06	< 2.2e-16 ***
ln_neer	-0.4955	0.0780	-6.514	5.10e-10 ***
policy_rate	0.0448	0.0030	14.751	< 2.2e-16 ***
vix	0.0070	0.0047	1.497	0.1350
post_crisis	-0.0394	0.1414	-0.279	0.7806

A.2 Fixed Effects — Full Output

Balanced Panel: n = 12, T = 39, N = 468 | $R^2 = 0.916$ | F-statistic: 1224.76 ($p < 2.2e-16$)

Variable	Estimate	Std. Error	t-value	Pr(> t)
ln_neer	-1.5550	0.0342	-45.454	< 2.2e-16 ***
policy_rate	0.0031	0.0005	6.205	3.60e-09 ***
vix	0.0030	0.0004	6.851	2.41e-11 ***
post_crisis	-0.0003	0.0133	-0.022	0.9826

A.3 Random Effects — Full Output (Preferred Model)

Balanced Panel: n = 12, T = 39, N = 468 | $R^2 = 0.913$ | Chi-sq: 4867.45 ($p < 2.2e-16$)

Variable	Estimate	Std. Error	z-value	Pr(> z)
(Intercept)	12.0438	0.1880	64.068	< 2.2e-16 ***
ln_neer	-1.5516	0.0343	-45.297	< 2.2e-16 ***
policy_rate	0.0032	0.0005	6.078	1.22e-09 ***
vix	0.0030	0.0004	6.818	9.23e-12 ***
post_crisis	-0.0003	0.0134	-0.022	0.9824

A.4 Robust Standard Errors (Arellano HC3) — Fixed Effects

Variable	Estimate	Robust SE	t-value	Pr(> t)
ln_neer	-1.5550	0.1026	-15.158	< 2.2e-16 ***
policy_rate	0.0031	0.0009	3.376	0.000798 ***
vix	0.0030	0.0006	4.713	3.25e-06 ***
post_crisis	-0.0003	0.0137	-0.021	0.983081

A.5 Heterogeneity Analysis: Importers vs. Exporters

Variable	Importers (n=8, N=312, R ² =0.929)	Exporters (n=4, N=156, R ² =0.811)
ln_neer	-1.563*** (0.038)	-1.518*** (0.104)
policy_rate	0.004*** (0.001)	-0.001 (0.001)
vix	0.004*** (0.001)	0.002*** (0.001)
post_crisis	-0.021 (0.018)	0.039** (0.017)